

Name	Amogh G. Okade
Roll number	EP22B020

2.1:

classmate
Date _____
Page _____

6.2.26 Assignment - 2

A.2.1) a) $f = \frac{1}{2\pi R_0 C_0} \Rightarrow 10^7 = \frac{1}{2\pi R_0 C_0}$

$f_{\min} = \frac{1}{2\pi \times 1.25 R_0 \times 1.15 C_0} = 10^7 = 6.956 \text{ MHz}$

$f_{\max} = \frac{1}{2\pi \times 0.75 R_0 \times 0.85 C_0} = 10^7 = 15.686 \text{ MHz}$

b) If $C = 0.85 \times 10^{-11} \text{ F}$, to prevent f from exceeding $f_{\max}$, I can have the maximum possible resistance, $R_x \pm 0.25 R_x$. Since it has to work for all variations of R_x ,

$$f_{\max} = 10.25 \times 10^6 > \frac{1}{2\pi \times 0.75 R_x \times 0.85 \times 10^{-11}}$$

$$\therefore R_x > \frac{10^5}{2\pi \times 0.75 \times 0.85 \times 10.25} = 2.4356 \text{ k}\Omega$$

→ If $C = 1.15 \times 10^{-11} \text{ F}$, to prevent f from dipping below $f_{\min}$, I can decrease my nominal resistance to $R_x \parallel R_y / N$. The worst-case minimum resistance will then be $1.25 \times (R_x \parallel R_y / N)$

$$\therefore f_{\min} = 9.75 \times 10^6 < 1$$

$$\therefore \frac{a_y}{N} < \frac{10^5 a_x}{2\pi \times 1.15 \times 1.25 \times 9.75 a_x - 10^5} = 2.128 \times 10^3$$

→ For some variation d , in C and d_z in a_x , say the bandwidth fell below $f_{\min}$.

$$\therefore \frac{1}{2\pi d, C \quad d_z a_x} < f_{\min} = 9.75 \times 10^6$$

$$\therefore d, d_z > \frac{10^5}{2\pi \times 9.75 \times a_x} = 0.67$$

Here, we would have to switch ON an a_y , which has $0.75 a_y$ value for the worst case. Then, the new bandwidth should not exceed $f_{\min}$.

$$\therefore 10.25 \times 10^6 > \frac{1}{2\pi d, C \times (d_z a_x \parallel 0.75 a_y)}$$

$$\text{Solving, } a_y > \frac{10^5 d_z a_x}{2\pi \times 0.75 \times 10.25 \times d a_x - 0.75 \times 10^5}$$

For the strongest constraints, we use the minimum d and the maximum d_z .

$$d_{\min} \approx 0.67$$

classmate
 Date _____
 Page _____

$$\alpha_{2 \max} = \frac{\alpha}{\alpha_{1 \min}} = \frac{0.67}{0.85} = 0.788$$

$$\therefore R_y > 50.463 \text{ k}\Omega$$

— $N > \frac{R_y}{2.128 \times 10^3} = 23.714$

$$\therefore N_{\min} = 24 \text{ switches}$$

* To confirm that this is indeed the optimal answer, we would have to check the equations for the cases when each R_y is switched ON.

2.2:

classmate
Date _____
Page _____

A2.2 a) $f = \frac{1}{2\pi\alpha C} \Rightarrow 10^7 = \frac{1}{2\pi [\alpha_p(f) + \alpha_n(f)] \times 10^{-11}}$

$\therefore \alpha_p(f) + \alpha_n(f) = 10^4 / 2\pi$ ✓

$\alpha_p(f_0) + \alpha_n(f_0) + (f - f_0) \times 10^{-3} [\alpha_p(f_0) - 2\alpha_n(f_0)] = \frac{10^4}{2\pi}$ — (1)

$f = f_0 \therefore \alpha_p(f_0) + \alpha_n(f_0) = 10^4 / 2\pi$ — (2)

From (1) and (2),

$(f - f_0) \times 10^{-3} [\alpha_p(f_0) - 2\alpha_n(f_0)] = 0$ ✓

$\therefore \alpha_p(f_0) = 2\alpha_n(f_0)$ — (3)

From (2) and (3),

$\alpha_n(f_0) = \frac{10^4}{6\pi} = 530.51 \Omega \quad \alpha_p(f_0) = 1061.033 \Omega$

$\therefore (L/W)_n = \frac{530.51}{150} = 3.537$ squares $\quad (L/W)_p = 10.61$ squares

b) $\alpha_p(27^\circ) = \alpha_p(f_0) = \frac{1}{2\pi \times 10^3 C} = \frac{10^4}{2\pi}$

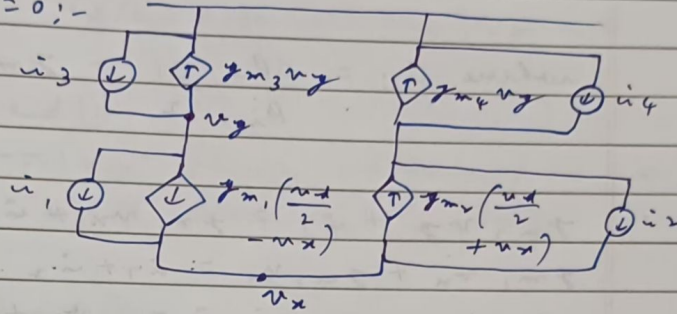
$\therefore \alpha_p(f) = \frac{10^4}{2\pi} [1 + 10^{-3} (f - 27^\circ)]$

$\alpha_p(100^\circ) = \frac{10^4}{2\pi} [1 + 10^{-3} \times 73^\circ] \quad \alpha_p(-40^\circ) = \frac{10^4}{2\pi} [1 - 10^{-3} \times 67^\circ]$

$f_{min} = \frac{1}{2\pi \alpha_p(100^\circ) \times C} = 9.32 \text{ MHz} \quad f_{max} = \frac{1}{2\pi \alpha_p(-40^\circ) \times C} = 10.718 \text{ MHz}$

where $A^2_{\beta N} = W_1 L_1 \sigma^2 (0\beta_1/\beta_1) = W_2 L_2 \sigma^2 (0\beta_2/\beta_2)$
 $A^2_{\beta \phi} = W_3 L_3 \sigma^2 (0\beta_3/\beta_3) = W_4 L_4 \sigma^2 (0\beta_4/\beta_4)$
 and $A^2_{VTN} = W_1 L_1 \sigma^2 (0V_{T1}) = W_2 L_2 \sigma^2 (0V_{T2})$
 $A^2_{VT\phi} = W_3 L_3 \sigma^2 (0V_{T3}) = W_4 L_4 \sigma^2 (0V_{T4})$

c) For $I_D = 0$:-



$$g_{m3} v_g + g_{m1} (v_d/2) - g_{m1} v_x + i_1 = i_3$$

$$i_1 + g_{m1} (v_d/2) - g_{m1} v_x + i_2 = g_{m2} v_d/2 + g_{m2} v_x$$

$$g_{m4} v_g + i_2 = i_4 + g_{m2} (v_d/2) + g_{m2} v_x$$

$$\therefore v_d = \frac{(g_{m3} + g_{m4})(g_{m1} i_2 - g_{m2} i_1) + (g_{m1} + g_{m2})(g_{m4} i_3 - g_{m3} i_4)}{g_{m1} g_{m2} (g_{m3} + g_{m4})}$$

$$d) \sigma^2(v_d) = \left(\frac{I_1}{2}\right)^2 \left[(g_{m3} + g_{m4})^2 \left(\frac{g_{m1}^2}{W_1 L_1} + \frac{g_{m2}^2}{W_2 L_2} \right) A^2_{\beta N} \right. \\ \left. + (g_{m1} + g_{m2})^2 \left(\frac{g_{m4}^2}{W_3 L_3} + \frac{g_{m3}^2}{W_4 L_4} \right) A^2_{\beta \phi} \right] \\ + (g_{m3} + g_{m4})^2 (g_{m1} g_{m2})^2 (1/W_1 L_1 - 1/W_2 L_2) A^2_{VTN} \\ + (g_{m1} + g_{m2})^2 (g_{m3} g_{m4})^2 (1/W_4 L_4 - 1/W_3 L_3) A^2_{VT\phi} \\ (g_{m1} g_{m2})^2 (g_{m3} + g_{m4})^2$$

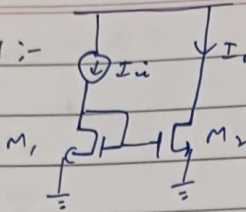
2.4:

CLASSMATE

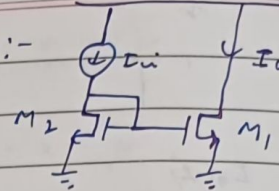
Date _____

Page _____

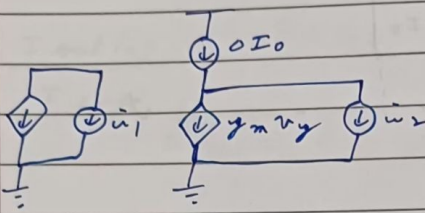
A 2.4 $S=1$:-



$S=0$:-



a)



$$\Delta I_o = \dot{u}_x - \dot{u}_y$$

$$= I_{i0} \left(\frac{\beta_2}{\beta_1} - \frac{\beta_1}{\beta_2} \right) + g_m (\Delta V_{T1} - \Delta V_{T2})$$

$$\therefore \sigma^2(\Delta I_o / I_{i0}) = 2\sigma^2(\beta) + g_m^2 / I_{i0}^2 \times 2\sigma^2(V_T) \quad - (1)$$

where $\sigma(\Delta I_o / I_{i0})$ is the S.D. of I_o / I_{i0}

b)

$$g_m = \frac{\Delta I_o}{V_{GS} - V_T} = 5 \times 10^{-3} \text{ S}$$

$$5 \times 10^{-4} = \frac{1}{2} \times 4 \times 10^{-4} \times \frac{W}{L} (2 \times 10^{-1})^2 \Rightarrow \frac{W}{L} = \frac{125}{2} \quad - (2)$$

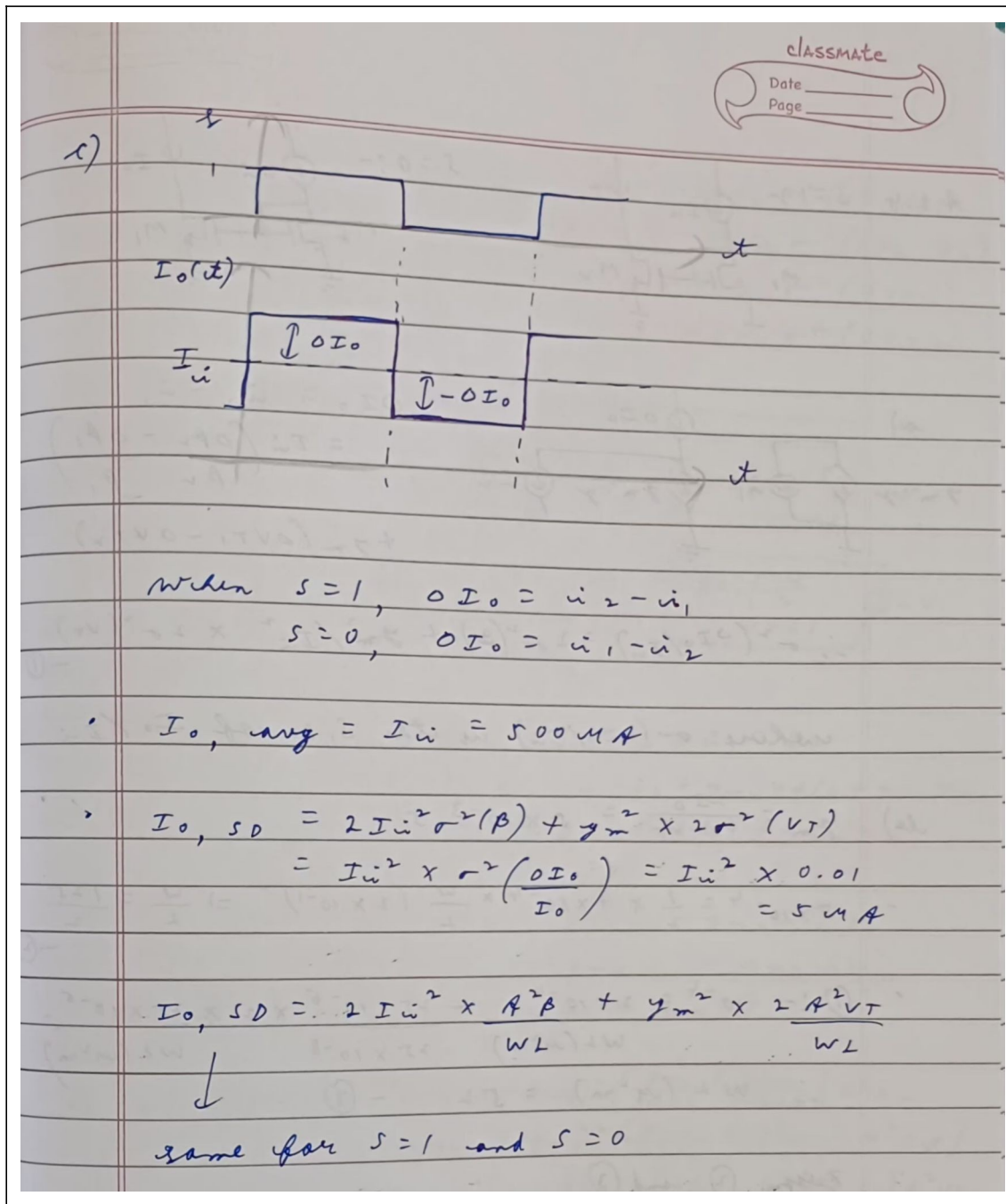
$$\textcircled{1} \text{ :- } 10^{-4} = \frac{2 \times 10^{-4}}{WL (\mu^2 \text{m})} + \frac{25 \times 10^{-6}}{25 \times 10^{-8}} \times 2 \times \frac{25 \times 10^{-6}}{WL (\mu^2 \text{m})}$$

$$\therefore WL (\mu^2 \text{m}) = 52 \quad - (3)$$

From (2) and (3),

$$L = 0.912 \mu \text{m}$$

$$W = 57.008 \mu \text{m}$$



2.5:

classmate
Date _____
Page _____

A 2.5
$$I_{avg} = \frac{NI_0 + \Delta I_1 + \Delta I_2 + \dots + \Delta I_N}{N}$$

$$\frac{1+x}{1+y} \approx (1+x)(1-y) \approx 1+x-y$$

a)
$$\frac{I_{out}(n)}{I_{avg}} = N \left[\frac{nI_0 + \Delta I_1 + \dots + \Delta I_n}{NI_0 + \Delta I_1 + \dots + \Delta I_N} \right]$$

$$= n \left[1 + \frac{(\Delta I_1 + \dots + \Delta I_n)}{I_0} \left(\frac{1}{n} - \frac{1}{N} \right) - \frac{(\Delta I_{n+1} + \dots + \Delta I_N)}{NI_0} \right]$$

$$\left\langle \frac{I_{out}(n)}{I_{avg}} \right\rangle = n [1 + 0] = n \quad (\because \langle \Delta I_i \rangle = 0)$$

$$var = n^2 \left[0 + n \left(\frac{1}{n} - \frac{1}{N} \right)^2 - \frac{N-n}{N^2} \right] \sigma^2(\Delta I_0/I_0)$$

$$var = \sigma^2(\Delta I_0/I_0) (n - n^2/N)$$

$$\therefore SD = \sigma^2(\Delta I_0/I_0) \sqrt{n - n^2/N}$$

$$SD_{max} \Rightarrow var_{max} \Rightarrow \frac{d}{dn} \left(n - \frac{n^2}{N} \right) = 0$$

$$\therefore (1 - 2n/N) = 0 \Rightarrow n = N/2$$

$$\therefore SD_{max} = \sigma(\Delta I_0/I_0) \sqrt{N/2 - N/4} = \frac{\sqrt{N}}{2} \sigma(\Delta I_0/I_0)$$

b)
$$\frac{I_{out}(n+1) - I_{out}(n)}{I_{avg}} = N \left[\frac{I_0 + \Delta I_{n+1}}{NI_0 + \Delta I_1 + \dots + \Delta I_N} \right]$$

$$= 1 + \frac{\Delta I_{n+1}}{I_0} - \frac{(\Delta I_1 + \dots + \Delta I_N)}{NI_0}$$

classmate
Date _____
Page _____

$$= 1 - \frac{(OI_1 + \dots + OI_n + OI_{n+1} + \dots + OI_N) - OI_{n+1} \left(\frac{1}{NI_0} - 1 \right)}{NI_0}$$

- $\left(\frac{I_0[n+1] - I_0[n]}{I_{avg}} \right) = 1 \quad (\because \langle OI_i \rangle = 0)$
- $var = \frac{(N-1)}{N^2} \sigma^2(OI_0/I_0) + \left(\frac{1}{N} - 1 \right)^2 \sigma^2(OI_0/I_0)$
- $\therefore SD = \sigma(OI_0/I_0) \times \sqrt{\frac{1-1}{N}}$

1) $\frac{I_0[n+1] - I_0[n]}{I_{avg}} = \frac{N \left[I_0 + OI_{n+1} + \dots + OI_{n+1} - (OI_n + \dots + OI_1) \right]}{NI_0 + OI_1 + \dots + OI_n}$

$$= 1 + \frac{OI_{n+1} + \dots + OI_{n+1} - (OI_n + \dots + OI_1) - (OI_1 + \dots + OI_n)}{NI_0}$$

- $Mean = 1, \quad \because \langle OI_i \rangle = 0 \quad \forall i$
- $var = \sigma^2(OI_0/I_0) \left[\frac{2n+1-1}{N} \right] \Rightarrow SD = \sigma(OI_0/I_0) \sqrt{\frac{2n+1-1}{N}}$

where $\sigma^2(OI_0/I_0) = \sigma^2(OP/\beta) + g_m^2/I_0^2 \sigma^2(OVT)$

* In b), all OI_i (except OI_{n+1}) directly cancel out. So, there is no dependence on n . But in c), the errors may not cancel out; they may add up also. So, the SD in c) is higher than b) (and depends on n).

2.6:

classmate
Date _____
Page _____

A 2.6 a) $\dot{u}_i$:-

$$\dot{u}_o = -g_m v_x \quad \dot{u}_i = -v_x (g_m + sC)$$

$$\therefore \frac{\dot{u}_o}{\dot{u}_i} = \frac{g_m}{g_m + sC}$$

v_s :-

$$g_m(v_x - v_s) + (v_x - v_s) sC = 0$$

$$\therefore v_x = v_s$$

$$\therefore \dot{u}_o = 0$$

$$\therefore \frac{\dot{u}_o}{v_s} = 0$$

b) $\dot{u}_i$:-

$$\frac{\dot{u}_o}{\dot{u}_i} = \frac{g_m}{g_m + sC}$$

classmate

Date: _____

Page: _____

$v_s :-$

$$i_o = -g_m(v_x - v_s)$$

$$g_m(v_x - v_s) + v_x \times sC = 0$$

$$\therefore v_x = \frac{g_m v_s}{g_m + sC}$$

$$\therefore \frac{i_o}{v_s} = \frac{g_m sC}{g_m + sC}$$

* The current gain in both cases is the same, because the small signal equivalents are the same in both cases.

- When v_s is applied in (a), no excess current can flow through the PMOS and thus, 0 current is mirrored incrementally to the output.
- When the same is done in (b), the series combination of $\frac{1}{g_m}$ and $\frac{1}{sC}$ causes some current in the left PMOS, which is then mirrored to the right PMOS as output incremental current.
- So, we can say that the Bulk of PMOSs should be connected to VDD and not GND, so that changes in VDD do not affect the current mirrors due to the effect of C_{gs} .